

Simple Prescaler for Transmitters

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Most phase-locked loop (PLL) based transmitters use a prescaler chip at the heart of their oscillator. A single-chip prescaler like SAB6456 is a divide-by-64 or divide-by-256 switchable prescaler. It offers high sensitivity and ease of use. But electronics hobbyists, particularly in India, have a tough time finding these chips in the market.

The circuit presented here solves this problem by using three low-cost, readily-available integrated circuits (ICs) and offering three common divider rates of 64, 128 and 256. It works well from 1MHz to 150MHz and is suitable for use in PLL FM transmitters and other frequency

synthesisers for HAM band including 2-metre amateur radio band (144MHz-148MHz).

Circuit and working

Circuit diagram of the simple prescaler for transmitters is shown in Fig 1. It is built around

voltage regulator 7805 (IC1), dual D-type positive-edge-triggered flip-flop 74AC74 (IC2 and IC3), ripple-carry binary counter CD4040 (IC4) and a few other components.

The four cascaded flip-flops in IC2 and IC3 have been configured to divide the input signal by a factor of 16. Radio frequency (RF) input (at JACK1) from the oscillator is applied to pin 3 of IC2 via capacitor C3. Signal from IC2 pin 9 is fed to pin 3 of IC3. Transistor T1 converts TTL output of IC3 to CMOS logic levels to drive clock input to pin 10 of binary counter IC4.

Output divider rates of 64, 128 and 256 are available at pins 1, 2 and 3 of CON2, respectively. The circuit can directly drive a CMOS divider chip like CD4059 (not shown here). LED1 indicates that the prescaler is working normally. In the absence of RF input, LED1 remains off.

Construction and testing

An actual-size, single-side PCB pattern of the simple prescaler for transmitters is shown in Fig. 2 and its component layout in Fig. 3.

Test Points

Test point	Details
TP0	GND
TP1	12V DC
TP2	5V
TP3	RF input frequency /4
TP4	RF input frequency /16
TP5	RF input frequency /32
TP6	RF input frequency /64
TP7	RF input frequency /128
TP8	RF input frequency /256

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2, IC3 - 74AC74 dual D flip-flop
- IC4 - CD4040 ripple counter
- T1, T2 - 2N2222 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R2, R5, R7 - 10-kilo-ohm
- R3, R4 - 220-kilo-ohm
- R6 - 3.3-kilo-ohm

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2, C4, C5 - 0.1 μ F ceramic disk
- C3 - 10pF ceramic disk

Miscellaneous:

- CON1 - 2-pin connector terminal
- CON2 - 4-pin connector
- JACK1 - RF jack interface connector
- 2-pin connector

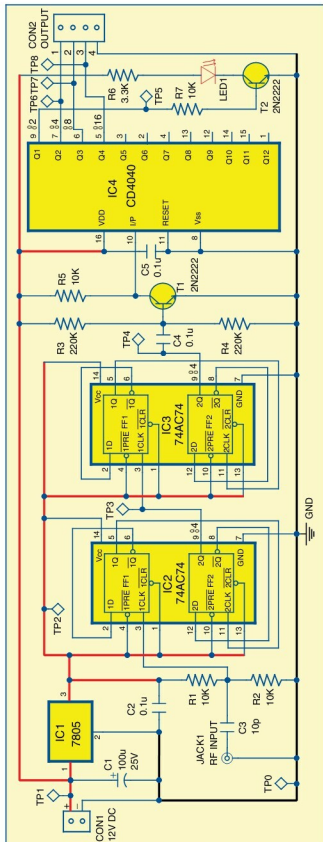


Fig. 1: Circuit diagram of the simple prescaler for transmitters